

Differentiate your Objective

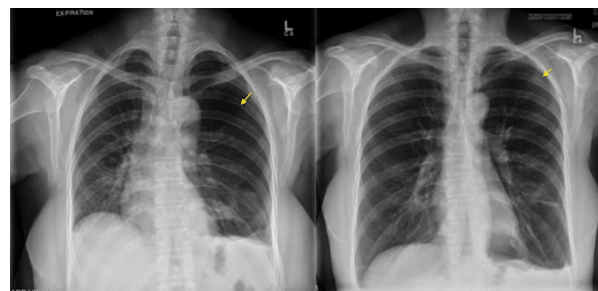
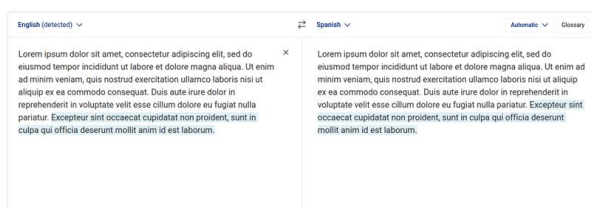
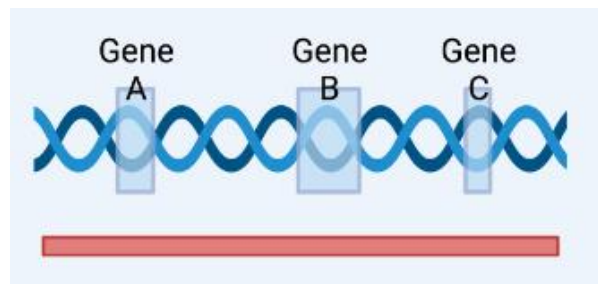
VLC = $\int \int \int$ Vision
Learning & Control

AICE3002 Introduction to Deep Learning
AICE6001 Introduction to Deep Learning (MSc)

Jonathon Hare and Antonia Marcu

Vision, Learning and Control
University of Southampton

What does Deep Learning (and Differentiable Programming) give us?



Machine Learning in a Nutshell

All credit for this slide goes to Niranjana

Data $\{\mathbf{x}_n, \mathbf{y}_n\}_{n=1}^N \quad \{\mathbf{x}_n\}_{n=1}^N$

Function Approximator $\mathbf{y} = f(\mathbf{x}; \boldsymbol{\theta})$

Loss function $\mathcal{L} = \sum_{n=1}^N \|\mathbf{y}_n - f(\mathbf{x}_n; \boldsymbol{\theta})\|_2^2$

Parameter Estimation $\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}} \mathcal{L}$

Prediction $\hat{\mathbf{y}}_{N+1} = f(\mathbf{x}_{N+1}, \hat{\boldsymbol{\theta}})$

What is Deep Learning?

Deep learning is primarily characterised by function compositions:

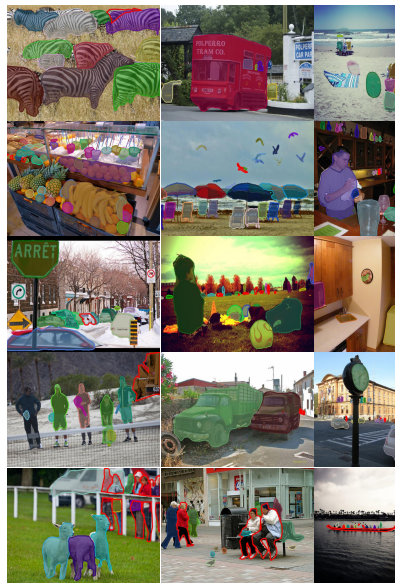
- Feedforward networks: $\mathbf{y} = f(g(\mathbf{x}; \boldsymbol{\theta}_g); \boldsymbol{\theta}_f)$
 - Often with relatively simple functions (e.g. $f(\mathbf{x}; \boldsymbol{\theta}_f) = \sigma(\mathbf{x}^\top \boldsymbol{\theta}_f)$)
- Recurrent networks:
 $\mathbf{y}_t = f(\mathbf{y}_{t-1}, \mathbf{x}_t; \boldsymbol{\theta}) = f(f(\mathbf{y}_{t-2}, \mathbf{x}_{t-1}; \boldsymbol{\theta}), \mathbf{x}_t; \boldsymbol{\theta}) = \dots$

What is Differentiable Programming?

- Captures the idea that computer programs can be partially learned rather than fully programmed by humans.
- The program is made of functions with optimisable parameters.
- So we need to be able to compute gradients with respect to the parameters of these functional blocks.
- The functional blocks don't need to be direct functions in a mathematical sense; more generally they can be *algorithms*.
- What if the functional block we're learning parameters for is itself an algorithm that optimises the parameters of an internal algorithm using a gradient based optimiser?!¹

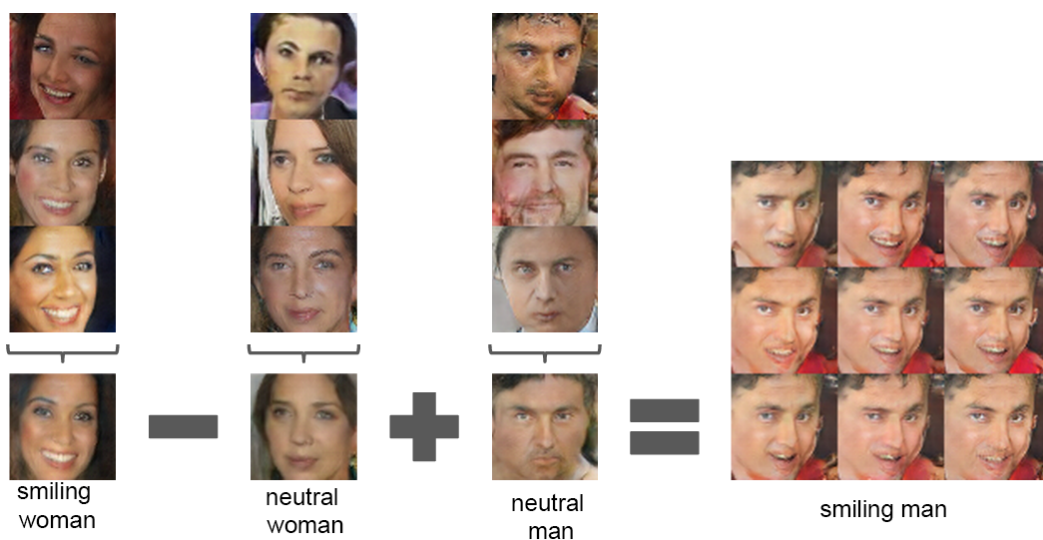
¹See our ICLR 2019 paper: <https://arxiv.org/abs/1812.03928> and NeurIPS 2019 paper: <https://arxiv.org/abs/1906.06565>

Success stories - Object detection and segmentation



Pinheiro, Pedro O., et al. "Learning to refine object segments." European Conference on Computer Vision. Springer, 2016.

Success stories - Image generation



Radford, Alec, Luke Metz, and Soumith Chintala. "Unsupervised representation learning with deep convolutional generative adversarial networks." arXiv preprint arXiv:1511.06434 (2015).

GPT-4

Prompt:

Translate the following Irish text to English:

"Bhí mé ag siúl sa choill nuair a chuala mé ceol álainn ón adharc."

Output (With LLM Integration):

I was walking in the forest when I heard beautiful music from the horn.

Output (Traditional MT):

I was in the wood when I heard nice sound from the horn.

Lyu, C., Du, Z., Xu, J., Duan, Y., Wu, M., Lynn, T., Aji, A.F., Wong, D.F., Liu, S. and Wang, L., 2023. A Paradigm Shift: The Future of Machine Translation Lies with Large Language Models. LREC-COLING 2024

Why should we care about this?

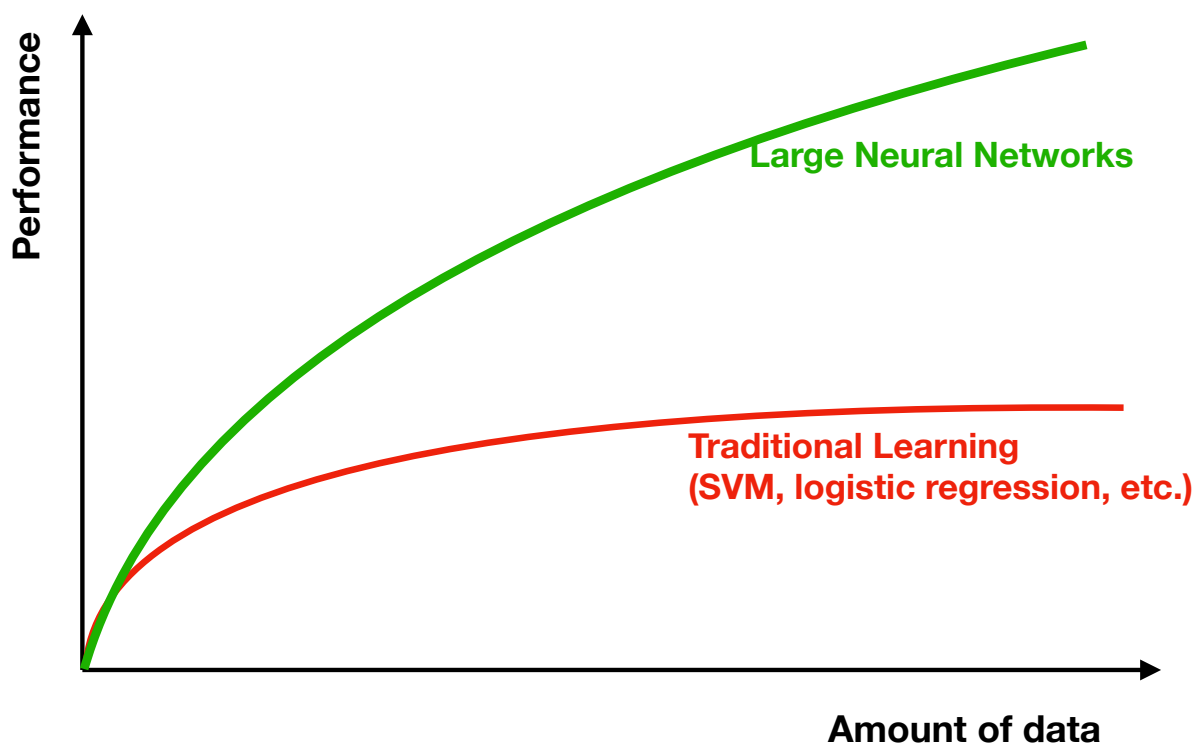


Image courtesy of Andrew Ng

Key takeaways

- **More data availability** in all domains...
- ...allows for **bigger/deeper/more complex models**
- ...models that can **learn the features**
- and these can have far **greater performance**

What is the objective of this module?

A word of warning: This is not a module about how to apply someone else's deep network architecture to a task, or how to train existing models!

You will learn some of that along the way of course, but the real objective is for you to graduate understanding how deep learning works beneath the hood.

What is the objective of this module?

- Understand the underlying mathematical and algorithmic principles of deep learning.
- Apply deep learning methods to real datasets.
- Learn how to train models using deep learning libraries.
- Think critically about why certain approaches were designed in the way they were.

How is this module going to be delivered?

- Lectures and seminars (3 most weeks)
 - This is a new module; we're utilising some material from our older module, but also adding/expanding and changing. Expect the moodle pages to change!
 - You need to read the suggested papers/links before the lectures!
 - There is maybe a little room for some flexibility later in the course on topics - tell us what you're interested in!
 - Lectures will be face to face, but also recorded for the website when possible (do not rely on this!).

How is this module going to be delivered?

- Seminars

- We'll look at a particular paper, papers or ideas in some detail and have an open discussion
- You'll need to prepare in advance & be ready to ask questions and share your thoughts
- Not recorded, but contents examinable

How is this module going to be delivered?

- Labs (1x 2 hour session per week for 8 weeks + one additional optional help / catchup session)
 - Labs consist of a number of Jupyter notebooks you will work through.
 - You'll be using PyTorch as the primary framework, with Torchbearer to help out.
 - You will need to utilise GPU-compute for the later labs (we provide Google Colab links so you can use NVidia GPUs in the cloud).
 - Labs are in-person (AICE lab in B32) with a team of PhD student demonstrators & both of us.
 - Please ask lots of questions and use this time to get help on the labs.
 - After each block of labs you will have to do a follow-up "Lab Investigation" that will be marked.

WEEK PLANNER						2026-27
Modules: AICE3002 / AICE6001						
Term	Academic	Monday	Tuesday	Wednesday	Thursday	Friday
Week	Week	21-Sep	22-Sep	23-Sep	24-Sep	25-Sep
1	1		09:00 07/3027 (L/R F1) Introduction		17:00 07/3031 (L/R F2) Differentiable Programming	13:00 27/2001 (L/R 1) Vector/matrix calculus, Jacobians, JVPs, and VJP
2	2	28-Sep	09:00 07/3027 (L/R F1) Matrix completion, the Netflix Challenge, and data anonymity	30-Sep Lab 1: Tensors	1-Oct 17:00 07/3031 (L/R F2) Computing gradients I	2-Oct 13:00 27/2001 (L/R 1) Computing gradients II
3	3	5-Oct	09:00 07/3027 (L/R F1) Backpropagation	7-Oct Lab 2: Automatic Differentiation	8-Oct 17:00 07/3031 (L/R F2) Optimisation	9-Oct 13:00 27/2001 (L/R 1) Seminar: Memorisation
4	4	12-Oct	09:00 07/3027 (L/R F1) Fundamentals review	14-Oct Lab 3: Optimisation	15-Oct 17:00 07/3031 (L/R F2) Basic ML I	16-Oct 13:00 27/2001 (L/R 1) Basic ML II
5	5	19-Oct	09:00 07/3027 (L/R F1) Going deep	21-Oct Lab 4: Linear Models	22-Oct 17:00 07/3031 (L/R F2) Practical training	23-Oct lab 3&2 submission
6	6	26-Oct	09:00 07/3027 (L/R F1) Convolution fundamentals	28-Oct Lab 5: MLP	29-Oct 17:00 07/3031 (L/R F2) CNNs I	30-Oct 13:00 27/2001 (L/R 1) CNNs II
7	7	2-Nov	09:00 07/3027 (L/R F1) CNNs III	4-Nov Lab 6: CNN	5-Nov 17:00 07/3031 (L/R F2) Seminar: shape & texture	6-Nov 13:00 27/2001 (L/R 1) Embeddings
8	8	9-Nov	09:00 07/3027 (L/R F1) Seminar: transfer learning	11-Nov Lab 7: Transfer Learning	12-Nov 17:00 07/3031 (L/R F2) Tokenisation	13-Nov 13:00 27/2001 (L/R 1) RNNs
9	9	16-Nov	17-Nov XXX	18-Nov Consolidation Lab Dropin	19-Nov 17:00 07/3031 (L/R F2) AIMA	20-Nov lab 3-5 submission
10	10	23-Nov	09:00 07/3027 (L/R F1) LSTM	25-Nov Lab 8: Sequence Models	26-Nov 17:00 07/3031 (L/R F2) Attention	27-Nov 13:00 27/2001 (L/R 1) Transformers
11	11	30-Nov	1-Dec 09:00 07/3027 (L/R F1) Autoencoders	2-Dec	3-Dec XXX	4-Dec lab 6&7 submission
12	12	7-Dec	8-Dec 09:00 07/3027 (L/R F1) Revision	9-Dec	10-Dec XXX	11-Dec lab 8 submission
13, 14, 15		CHRISTMAS				
16, 17		SEMESTER 1 EXAMS				

What will we cover in the module?

AICE3002:

<https://moodle.ecs.soton.ac.uk/course/view.php?id=1031>

AICE6001:

<https://moodle.ecs.soton.ac.uk/course/view.php?id=1034>

Lab session plan

Lab	Date	Topic
Lab 1	30-Sep	Tensor programming with PyTorch
Lab 2	07-Oct	Automatic Differentiation
Lab 3	14-Oct	Optimisation
Lab 4	21-Oct	Linear Models
Lab 5	28-Oct	NNs with PyTorch and Torchbearer
	04-Nov	(Catch up / help)
Lab 6	11-Nov	CNNs with PyTorch and Torchbearer
Lab 7	18-Nov	Using pretrained models and transfer learning
Lab 8	25-Nov	Recurrent Networks, Sequence Prediction & Embeddings

Assessment Structure

- Lab Investigations 40%
- Final exam 60%

Note these are the same for both AICE3002 and AICE6001

Investigation	Submission Date	Labs covered
1 - Gradients and Matrix Completion	23rd Oct, 4PM	1 and 2
2 - Optimisation and Wide Networks	20th Nov, 4PM	3, 4 and 5
3 - Convolutions and Transfer	27th Nov, 4PM	6 and 7
4 -Transforming Sequences	11th Dec, 4PM	8

Exam

- 2 hour duration, in-person, written.
- Common Section A questions.
- Different Section B questions for AICE3002 and AICE6001.
- More details to follow later in the semester, including a cut-down sample paper with representative questions

What do we expect you already know?

- You have have taken Foundations of Machine Learning, or are currently taking it.
- Fundamentals of:
 - Matrix Algebra (matrix-matrix, matrix-vector and matrix-scalar operations, inverse, determinant, rank, Eigendecomposition, SVD);
 - Probability & Statistics (1st-order summary statistics, simple continuous and discrete probability distributions, expected values, etc); and,
 - Multivariable Calculus (partial differentiation, chain-rule).
- Programming in Python

What might you already know?

- How to use a deep learning framework (Keras, Tensorflow, PyTorch)?
- How to train an existing model architecture using a GPU?
- How to perform transfer learning?
- How to perform differentiable sampling of a Multivariate Normal Distribution?

What comes after this module?

- AICE6002: Deep Learning Research
 - taught by both of us
 - focus on translating the fundamentals you learn in this module to cutting edge research
 - plus lectures on advanced architectures, generative models, etc

Homework & questions

- Homework: Register for access to SWARM - <https://tinyurl.com/ecs-swarm>
- Any questions?